Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	15/10/2026
Team ID	
Project Name	Metro Ticket Booking System using ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Login	Registration through the portal form (name, email, password) Login through email and password / single sign-on
FR-2	Ticket Booking	Select source and destination station Select journey date, number of passengers and ticket type Submit request through the Service Catalog item
FR-3	Fare Calculation	Automatic fare lookup from the fare matrix Total fare = fare per passenger x number of passengers Fare shown to the user before confirmation
FR-4	Digital Ticket Generation	Unique ticket number generated automatically QR code generated and attached to the ticket record Ticket viewable and downloadable in My Tickets
FR-5	Notifications	Booking confirmation via Email with ticket details and QR code Journey reminder notification Cancellation notification
FR-6	Ticket Management	View booking history Cancel an unused ticket Re-book a previous journey
FR-7	Administration & Validation	Add / edit stations and fares Validate the ticket by scanning the QR code at the gate View booking and revenue reports

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The booking form is simple and can be completed in a few steps. The Service Portal is responsive on mobile and desktop, uses clear labels and drop-downs for stations, and supports accessible design (readable fonts, keyboard

NFR No.	Non-Functional Requirement	Description
		navigation).
NFR-2	Security	Role-based access control (passenger, gate staff, customer care, admin) using ServiceNow roles and ACLs. Passengers see only their own tickets. Data is transmitted over HTTPS, and QR codes carry a unique ticket reference that is verified against the database.
NFR-3	Reliability	Tickets are stored in ServiceNow tables with a status (Booked, Used, Cancelled). Flow executions are logged, so failed steps can be found and re-run. Each ticket number is unique.
NFR-4	Performance	Fare calculation and ticket creation run automatically in Flow Designer. Design target: booking confirmation and email within a few seconds under normal load.
NFR-5	Availability	The application runs on the ServiceNow cloud instance, so it is available 24x7 to passengers, apart from the platform's planned maintenance.
NFR-6	Scalability	New stations, routes and fares are added as data records without code changes. The platform can handle growth in users and bookings, and the same flow can support more lines or cities.